lecture17 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a program that reads a string the user and then prints the letter and index value in the style shown.

```
Enter a string: hello

s = 'hello'
s[0] is 'h'
s[1] is 'e'
s[2] is 'l'
s[3] is 'l'
s[4] is 'o'
```

Sample Solution

```
s = input('Enter a string: ')
print()
print(f"s = '{s}'")
for i in range(len(s)):
    letter = s[i]
    print(f"s[{i}] is '{letter}'")
```

Question 2 (q2_sol.py)

Write a program that repeatedly asks the user for a date in the format YYYY-MM-DD and prints the same date in the format DD/MM/YYYY. The program should end with the user pressing <enter> with no date entered.

You can assume the user always types the date in the correct format.

```
Enter a date (YYYY-MM-DD): 2026-10-25
25/10/2026
Enter a date (YYYY-MM-DD): 1992-08-14
14/08/1992
Enter a date (YYYY-MM-DD): <enter>
Done!
```

Sample Solution

```
while True:
    date = input('Enter a date (YYYY-MM-DD): ')
    if date == '':
        break
    year = date[0:4]
    month = date[5:7]
    day = date[8:10]
    print(f'{day}/{month}/{year}')
```

Question 3 (q3_sol.py)

Write a program that repeatedly asks the user to enter a word, and then prints the plural of the word (as described below). The program should stop when the user enters an empty string.

Write a function called pluralize(s) that takes a string s and returns a plural version of the string according to these rules:

- If the string ends with 's', do nothing.
- If the string ends with 'y', replace it with 'ies'.
- Otherwise, add an 's' to the end of the string.

Sample Run

```
Enter a word: cat
cats
Enter a word: baby
babies
Enter a word: penguins
penguins
Enter a word: <enter>
Done!
```

Sample Solution

```
def pluralize(s):
    if s[-1] == 's':
        return s
    elif s[-1] == 'y':
        return s[:-1] + 'ies' # remove the last letter and add 'ies'
    else:
        return s + 's' # add 's' to the end of the string

while True:
    word = input('Enter a word: ')
    if word == '':
        break
    print(pluralize(word))
print('Done!')
```

Question 4 (q4_sol.py)

Write a shopping list program where the user is asked to enter the name of a file they want to write to, and then repeatedly asks the user to enter grocery list items. Each item should be written to the file on its own line. The program should stop when the user presses <enter> without typing an item.

Sample Run

```
What file do you want to write to? list.txt
Enter an item: apples
Enter an item: bananas
Enter an item: carrots
Enter an item: <enter>

Done!
See list.txt for your list.
```

Sample Solution

```
filename = input('What file do you want to write to? ')
outfile = open(filename, 'w') # 'w' for writing to the file

item = input('Enter an item: ').strip()
while item != '':
    outfile.write(item + '\n') # \n ends a line in the file
    item = input('Enter an item: ').strip()
outfile.close()

print('Done!')
print(f'See {filename} for your list.')
```